

Project Implementation Report: Week 2

Project Title: Interactive Tic Tac Toe Application in Java

Operational Domain: Requirement Analysis, Game Rules Definition, and System Architecture Design

1. Requirement Analysis and Layout Planning

The primary objective of Week 2 was to analyze the operational requirements of the application and define the interaction flow between different interface components. During this phase, special attention was given to screen organization, user navigation, and gameplay transitions.

Key planning decisions included:

- Designing a dedicated Home Screen to serve as the initial landing page of the application.
- Creating a separate gameplay interface to host the active Tic Tac Toe board.
- Defining a navigation mechanism that allows seamless movement between the home interface and the game interface.
- Establishing clear visual separation between menu controls and gameplay components.

These design considerations ensured that the user interface would remain organized, intuitive, and easy to navigate.

2. Game Rules and Operational Logic

The second stage focused on formalizing the gameplay rules and internal operational behavior of the system.

The following gameplay rules were established:

- Player X always starts the game as the default first player.
- Turns alternate automatically between Player X and Player O after every valid move.
- A player cannot place a symbol in an already occupied grid position.
- The game immediately ends when a player completes a valid winning pattern.
- If all cells become occupied without a winning pattern, the game is declared a draw.
- A restart operation restores the board to its initial empty state.

These rules provide a structured and fair gameplay environment for both participants.

3. Screen Navigation Design

The application requires multiple interface screens that operate independently while sharing the same application window.

The navigation strategy consists of:

1. Home Screen
 - Displays the project title and game launch controls.
 - Serves as the application's entry point.
2. Game Screen
 - Contains the gameplay board and active player indicators.
 - Handles all in-game interactions and state updates.
3. Transition Mechanism
 - Transfers the user from the Home Screen to the Game Screen upon pressing the Start button.
 - Supports future scalability for additional screens if required.

This modular navigation model improves maintainability and future expansion capabilities.

4. MVC Architecture Design

To ensure clean code organization and better software maintainability, the project architecture was conceptually divided according to the Model-View-Controller (MVC) design principle.

Model Layer

Responsible for storing and managing application data such as:

- Current player information.
- Board state values.
- Game status variables.

View Layer

Responsible for graphical representation including:

- Home interface components.
- Game board layouts.
- Buttons, labels, and status indicators.

Controller Layer

Responsible for:

- Capturing user actions.
- Processing button click events.
- Executing game rules.
- Updating both interface components and internal states.

The MVC approach reduces system complexity and promotes modular software development practices.

5. Weekly Deliverables

By the end of Week 2, the following objectives were successfully completed:

- Functional requirements were finalized.
- Gameplay rules and operational constraints were defined.
- Screen navigation flow was established.
- Interface transition strategies were planned.
- MVC-based architectural design was completed.

These achievements provide a strong structural foundation for the upcoming implementation phase, where project setup, class initialization, and development environment configuration will begin in Week 3.